



**KLE Technological
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**Department of Electronics and Communication
Engineering**

Internship Project Report

on

Standard Cell Library Creation Using GPDK45nm

at

Center for Integrated Circuits & Systems(CICS)

Submitted by:

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01FE22BEC274

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KLE Technological University,
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2025-2026



DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING

CERTIFICATE

This is to certify that internship project entitled “**Standard Cell Library Creation Using GPDK45nm**” is a bonafide work carried out by **Raju M Barker (01FE22BEC274)**. The internship project report has been approved, as it satisfies the requirements for the internship work prescribed by the university curriculum for the BE (VIII Semester) in the Department of Electronics and Communication Engineering at KLE Technological University for the academic year 2025–2026.

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- 1.
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INTERNSHIP CERTIFICATE

This is to certify that Mr. **Raju M Barker** (USN: **01FE22BEC274**),
a student of **Electronics and Communication Engineering (ECE)**, KLE Technological University,
has successfully completed an internship on "**Custom Layout and Physical Verification**" at **CICS**
from **19 January 2026 to 30 May 2026**.

Project: Standard Cell Library Creation Using GPDK45nm.

Key Contributions

- **Custom Layout Design:** Designed and verified CMOS layouts and standard cells using Cadence Virtuoso, performing schematic capture, layout implementation, DRC, LVS, and Parasitic Extraction (PEX).
- **Physical Verification:** Developed and validated Calibre DRC rule decks for Metal, Via, and Contact layers using SVRF, ensuring compliance with foundry design rules and manufacturability requirements.
- **Technology Exploration:** Studied and analyzed SCL180nm, UMC180nm, and GPDK45nm Process Design Kits (PDKs), including technology files, layer mapping, LEF/CDL files, and CMOS device structures.
- **Technology Development:** Created a custom SCL180nm technology file by defining process layers, routing directions, display properties, via structures, and technology constraints for layout design.
- **Performance Analysis:** Conducted pre-layout and post-layout simulations to analyze rise time, fall time, propagation delay, dynamic power, leakage power, and the impact of layout parasitics on circuit performance.
- **EDA Tools & Methodologies:** Gained hands-on experience with Cadence Virtuoso, Cadence Assura, Siemens Calibre, and industry-standard VLSI physical design and verification flows.

Mr. **Raju M Barker** demonstrated strong technical competence, analytical thinking, and professionalism throughout the internship.

We appreciate his sincere efforts and wish him success in his future academic and professional endeavors.

Date: 1st June 2026

Place: Hubballi, Karnataka

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The role of the management of **Center for Integrated Circuits & Systems (CICS)** was vital, and I would like to sincerely thank them for providing me with all the necessary infrastructure, EDA tools, laboratory facilities, and technical support throughout the course of this internship and project work.

-Raju M Barker

ABSTRACT

This internship project focuses on the design, layout implementation, characterization, and physical verification of a standard cell library using GPDK45nm technology in Cadence Virtuoso. The work involves the development of multiple combinational standard cells including INV_X1, OAI21_X1, OAI211_X1, OAI22_X1, and OAI221_X1 following a predefined 9-track standard cell architecture and industry-standard layout guidelines. Each cell was designed at the schematic level, converted into symbol views, implemented as custom layouts, and verified through Design Rule Check (DRC), Layout Versus Schematic (LVS), and Parasitic Extraction (PEX) processes. The extracted parasitic views were used to perform post-layout simulations and characterize the electrical performance of each cell under realistic operating conditions. Important parameters such as rise time, fall time, propagation delay, average power consumption, leakage power, dynamic power, and cell area were measured and compared between pre-layout and post-layout simulations to evaluate the impact of layout parasitics. The results demonstrate that parasitic resistances and capacitances increase delay and power consumption while maintaining functional correctness of the designed cells. The project provides practical exposure to the complete full-custom VLSI design flow including schematic design, custom layout development, parasitic-aware simulation, timing characterization, power analysis, and physical verification. The developed standard cells form a reusable foundation for digital integrated circuit design and contribute toward the creation of a scalable standard cell library for future ASIC and VLSI applications.

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Chapter 1

Introduction

Standard cell libraries are the foundation of modern digital integrated circuit design. They consist of pre-designed, pre-characterized, and pre-verified logic cells that are repeatedly used during ASIC implementation. This project focuses on the creation and characterization of a standard cell library using GPDK45nm technology in **Cadence Virtuoso IC6.1.864b.500.32**. The developed library includes INV_X1, OAI21_X1, OAI211_X1, OAI22_X1, and OAI221_X1 standard cells implemented using a 9-track standard cell architecture. Each cell is designed following standard layout conventions including fixed cell height, power rail alignment, routing accessibility, and pin placement requirements. The design flow includes schematic design, symbol generation, and custom layout implementation using Cadence Virtuoso. Physical verification is performed using **Cadence Assura** through Design Rule Check (DRC), Layout Versus Schematic (LVS), and Parasitic Extraction (PEX) to ensure manufacturability and functional correctness. Pre-layout and post-layout simulations are performed to characterize timing, power, and area parameters and to study the impact of layout parasitics. This work establishes a complete standard cell development methodology and provides practical experience in custom VLSI design, physical verification, layout optimization, and standard cell characterization techniques used in the semiconductor industry.

1.1 Motivation

The motivation behind this project is to gain practical knowledge and hands-on experience in full-custom VLSI design and standard cell library development, which are essential components of modern semiconductor design flows. Standard cell libraries serve as the building blocks of digital systems, and their design requires a strong understanding of CMOS circuits, layout techniques, timing analysis, power optimization, and physical verification methodologies. Developing a standard cell library from schematic design to post-layout characterization provides an opportunity to understand the complete ASIC design flow used in semiconductor industries. The project also helps in understanding the effects of layout parasitics on circuit performance and the importance of DRC, LVS, and PEX verification in ensuring manufacturable and reliable designs. By designing and characterizing multiple standard cells using GPDK45nm technology, this work develops practical skills in Cadence Virtuoso and establishes a strong foundation for future work in digital VLSI design, physical design, standard cell development, and ASIC implementation.

1.2 Objectives and specifications

The primary objective of this project is to design, implement, characterize, and physically verify a standard cell library using GPDK45nm technology in Cadence Virtuoso. The specific objectives are:

- Design schematic implementations of INV_X1, OAI21_X1, OAI211_X1, OAI22_X1, and OAI221_X1 standard cells.
- Create reusable symbol views for integration into higher-level circuits and test-benches.
- Implement custom layouts following a 9-track standard cell architecture with a fixed cell height of 1.8 μm .
- Perform Design Rule Check (DRC) verification to ensure layout manufacturability.
- Perform Layout Versus Schematic (LVS) verification to confirm functional equivalence between schematic and layout.
- Generate extracted views using Parasitic Extraction (PEX).
- Perform pre-layout and post-layout simulations for timing and power characterization.
- Measure rise time, fall time, propagation delay, average power, leakage power, dynamic power, and cell area.
- Analyze the impact of parasitic effects on timing and power performance.
- Establish a reusable methodology for future standard cell library development.

1.3 Problem statement

The objective of this project is to develop a physically verified and fully characterized standard cell library using GPDK45nm technology in Cadence Virtuoso while adhering to predefined 9-track standard cell architecture guidelines. The designed cells must satisfy functional correctness, layout design rules, pin accessibility requirements, and routing constraints while achieving optimized performance in terms of propagation delay, power consumption, and area utilization. The project requires performing complete schematic design, custom layout implementation, DRC verification, LVS verification, parasitic extraction, and post-layout characterization for multiple standard cells. Furthermore, the impact of layout-induced parasitic effects on timing and power performance must be analyzed through detailed comparison of pre-layout and post-layout simulation results. The resulting standard cell library should provide reliable and reusable building blocks for future digital ASIC and VLSI design applications.

1.4 Standard Cell Characterization Methodology

The performance of the developed standard cells is evaluated using transient simulations in Cadence Virtuoso with a supply voltage of 1.1V. Important timing, power, and area parameters are calculated using the Calculator tool available in the Analog Design Environment (ADE). The following measurement definitions, equations, and calculator expressions are used for characterization of all standard cells developed in this work.

1.4.1 Rise Time

Rise time is defined as the time taken by the output signal to transition from 10% of VDD to 90% of VDD. For a supply voltage of 1.1V, the transition levels correspond to 0.11V and 0.99V respectively.

$$\text{Rise Time} = t_{90\%} - t_{10\%} \quad (1.1)$$

Calculator Expression:

```
riseTime(v("/Y" ?result "tran") 0 nil 1.1 nil 10 90 nil "time")
```

1.4.2 Fall Time

Fall time is defined as the time taken by the output signal to transition from 90% of VDD to 10% of VDD.

$$\text{Fall Time} = t_{10\%} - t_{90\%} \quad (1.2)$$

Calculator Expression:

```
fallTime(v("/Y" ?result "tran") 1.1 nil 0 nil 90 10 nil "time")
```

1.4.3 Rise Delay (t_{pLH})

Rise delay (t_{pLH}) is defined as the delay between the input signal falling through 50% of VDD and the output signal rising through 50% of VDD. For VDD = 1.1V, the 50% level corresponds to 0.55V.

$$t_{pLH} = t_{Output(50\%,rise)} - t_{Input(50\%,fall)} \quad (1.3)$$

Calculator Expression:

```
delay(v("/Input") v("/Output") 0.55 "falling" 0.55 "rising")
```

1.4.4 Fall Delay (t_{pHL})

Fall delay (t_{pHL}) is defined as the delay between the input signal rising through 50% of VDD and the output signal falling through 50% of VDD.

$$t_{pHL} = t_{Output(50\%,fall)} - t_{Input(50\%,rise)} \quad (1.4)$$

Calculator Expression:

```
delay(v("/Input") v("/Output") 0.55 "rising" 0.55 "falling")
```

1.4.5 Average Propagation Delay

The average propagation delay is calculated as the average of rise delay and fall delay.

$$t_p = \frac{t_{pLH} + t_{pHL}}{2} \quad (1.5)$$

1.4.6 Total Average Power

Total average power is defined as the average power drawn from the VDD supply during switching activity.

$$P_{avg} = VDD \times I_{avg} \quad (1.6)$$

Calculator Expression:

```
1.1*average(i("/VDD"))
```

1.4.7 Static / Leakage Power

Static or leakage power is measured when the input signals remain at stable logic levels and no switching activity occurs. Leakage measurements are performed for different valid input combinations of the respective standard cell.

$$P_{static} = VDD \times I_{leakage} \quad (1.7)$$

Calculator Expression:

```
1.1*average(i("/VDD"))
```

1.4.8 Dynamic Power

Dynamic power is calculated by subtracting the average static power from the total average power measured during switching operation.

$$P_{dynamic} = P_{total} - P_{static} \quad (1.8)$$

1.4.9 Cell Area

The cell area is measured from the layout using the PR boundary dimensions. The standard cell height is fixed at 1.8 μ m according to the adopted 9-track standard cell architecture, while the cell width varies depending on the complexity of the logic function.

$$Area = Width \times Height \quad (1.9)$$

The required measurements are performed using transient simulations with appropriate input stimulus, output load capacitance, and a supply voltage of 1.1V. Proper waveform and current saving options are enabled using the “Outputs $\rightarrow$ Save All” option in Cadence Virtuoso to ensure accurate extraction of timing and power parameters. The same characterization methodology is applied to all standard cells developed in this work.

Chapter 2

INV - X1

2.1 Cell Definition

Item	Specification
Cell name	INV_X1
Logic function	$Y = \overline{A}$
Drive strength	X1
Technology	GPDK45nm
Template	9T standard cell, 1.8 um height
Required pins	A, Y, VDD, VSS

2.2 Device Implementation

Device	Connection	Width	Length
PMOS	Source=VDD, Drain=Y, Gate=A, Body=VDD	1.35 um	45 nm
NMOS	Source=VSS, Drain=Y, Gate=A, Body=VSS	0.45 um	45 nm

2.3 Design Views Created

View	Requirement
schematic	CMOS inverter using the specified W/L and exact pin names
symbol	A, Y, VDD, VSS pins; clean testbench interface
layout	9T template layout; DRC/LVS clean
extracted av_extracted	/ Generated after parasitic extraction
testbench	Reusable for pre-layout and post-layout extracted simulations

2.3.1 Schematic

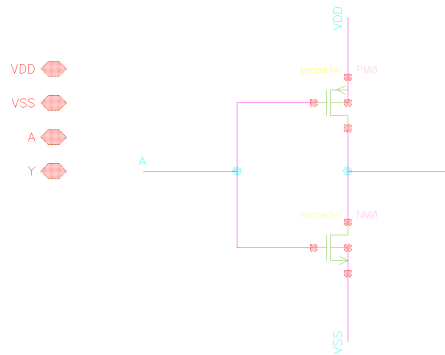


Figure 2.1: INV-X1-Schematic

The above figure shows the schematic design of the INV_X1 standard cell implemented using CMOS technology/ 33 in Cadence Virtuoso. The inverter consists of PMOS and NMOS transistors connected to provide complementary switching operation. The schematic verifies the basic inverter functionality with input A and output Y.

2.3.2 Symbol

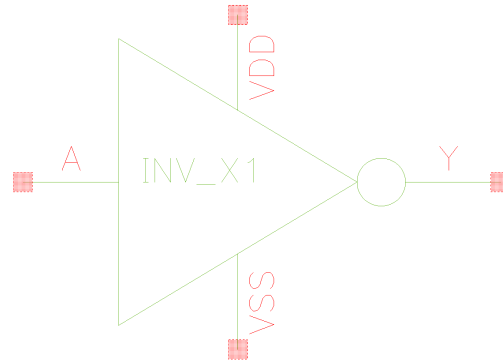


Figure 2.2: INV-X1-Symbol

The above figure shows the symbol view of the INV_X1 standard cell created in Cadence Virtuoso. The symbol contains input, output, and power pins required for integrating the inverter into higher-level circuit designs and testbenches.

2.3.3 Layout

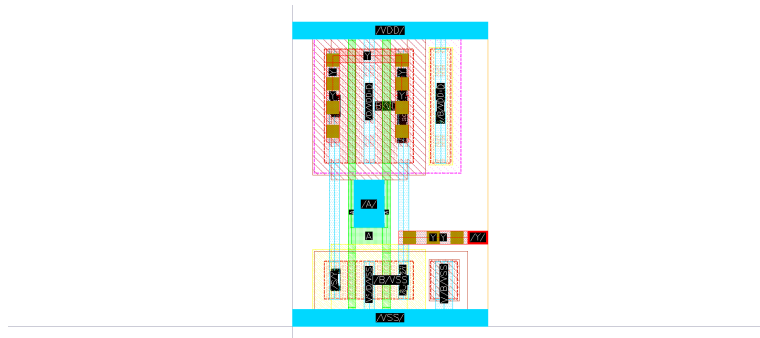


Figure 2.3: INV-X1-Layout

The above figure shows the layout implementation of the INV_X1 standard cell designed using the 9-track standard cell architecture in Cadence Virtuoso. The layout includes PMOS and NMOS transistors, power rails, routing connections, and pin accessibility while satisfying the required design rules and layout constraints.

2.3.4 AV - Extracted layout

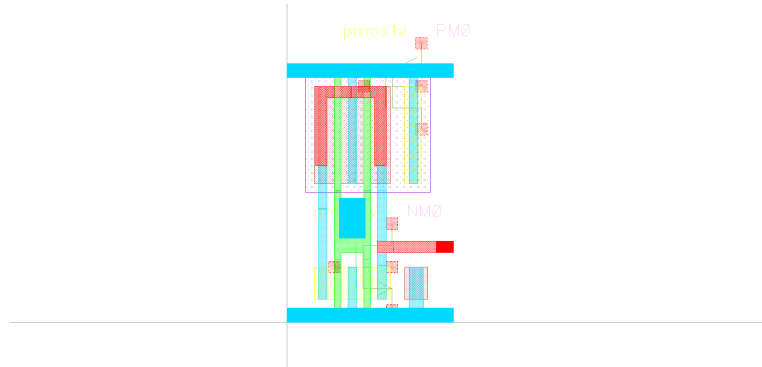


Figure 2.4: INV-X1-AV-Extracted

The above figure shows the av_extracted view of the INV_X1 standard cell generated after parasitic extraction. The extracted layout includes parasitic resistances and capacitances associated with the physical layout, enabling accurate post-layout simulation and performance analysis.

2.3.5 Testbench

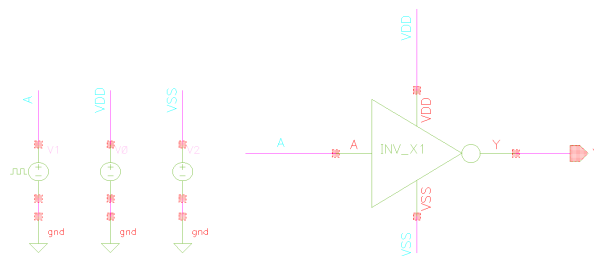


Figure 2.5: INV-X1-Testbench

The above figure shows the testbench setup used for simulation and characterization of the INV_X1 standard cell in Cadence Virtuoso. The testbench includes pulse input excitation, supply voltage, load capacitance, and measurement setup required for timing and power analysis.

2.4 Results

The performance of the INV_X1 standard cell is analyzed using transient simulations in Cadence Virtuoso with GPDK45nm technology. Important timing and power parameters such as rise time, fall time, rise delay, fall delay, and average power consumption are measured using the Cadence Calc/ 33ulator tool. The following waveforms and plots show the obtained simulation results for the designed inverter cell.

2.4.1 Rise Time Analysis

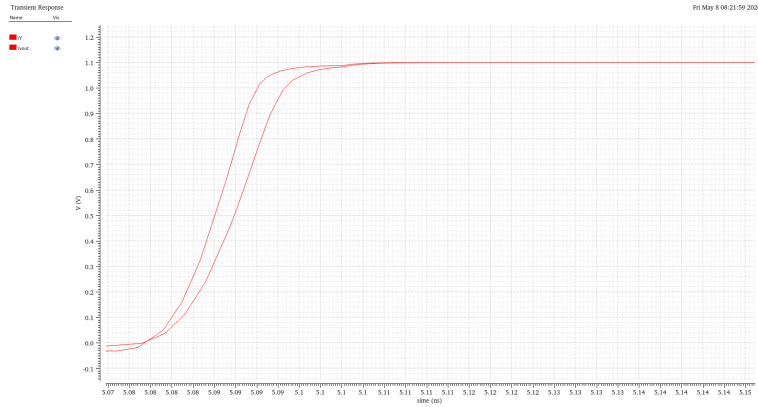


Figure 2.6: Rise Time Measurement of INV_X1

The above waveform shows the rise time measurement of the inverter output. The rise time is measured from 10% of VDD to 90% of VDD during the output transition from logic low to logic high.

2.4.2 Fall Time Analysis

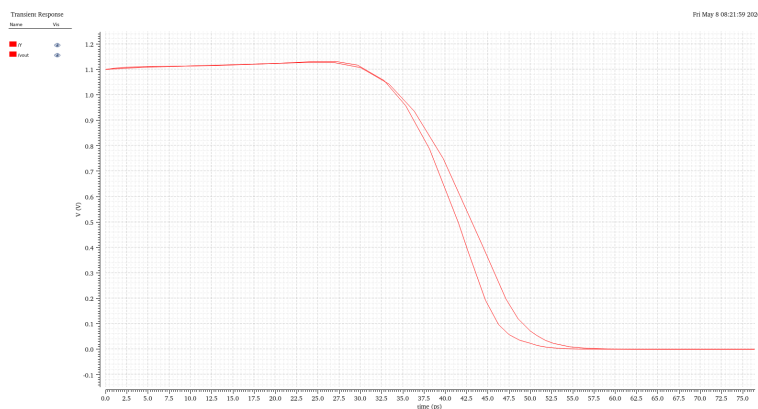


Figure 2.7: Fall Time Measurement of INV_X1

The above waveform shows the fall time measurement of the inverter output. The fall time is measured from 90% of VDD to 10% of VDD during the output transition from logic high to logic low.

2.4.3 Rise Delay (t_{pLH})

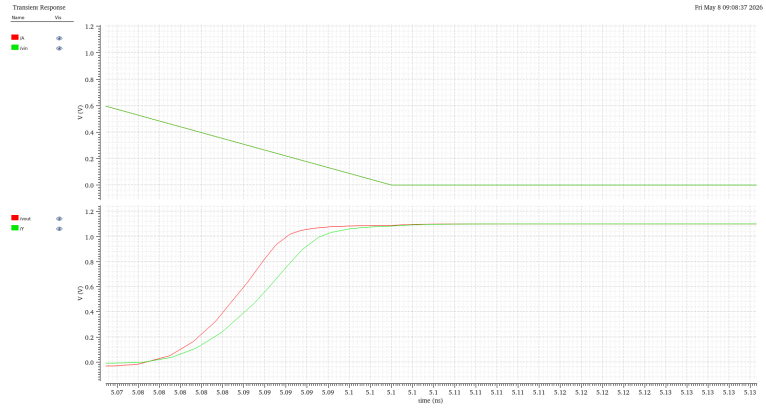


Figure 2.8: Rise Delay Measurement of INV_X1

The rise delay (t_{pLH}) represents the delay between the falling edge of the input signal and the rising edge of the output signal measured at 50% of VDD.

2.4.4 Fall Delay (t_{pHL})

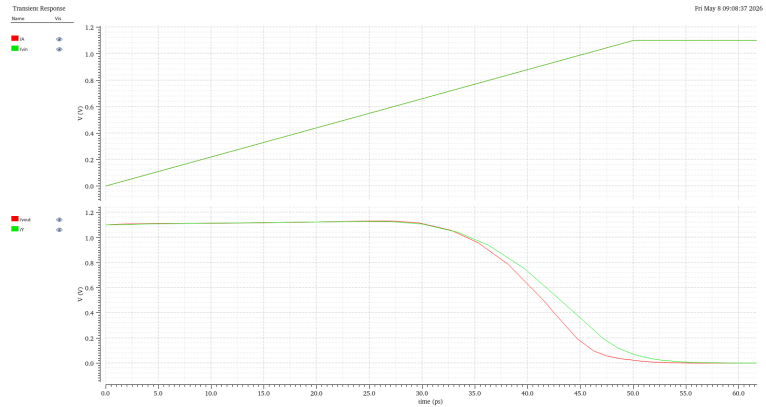


Figure 2.9: Fall Delay Measurement of INV_X1

The fall delay (t_{pHL}) represents the delay between the rising edge of the input signal and the falling edge of the output signal measured at 50% of VDD.

2.4.5 Power Analysis

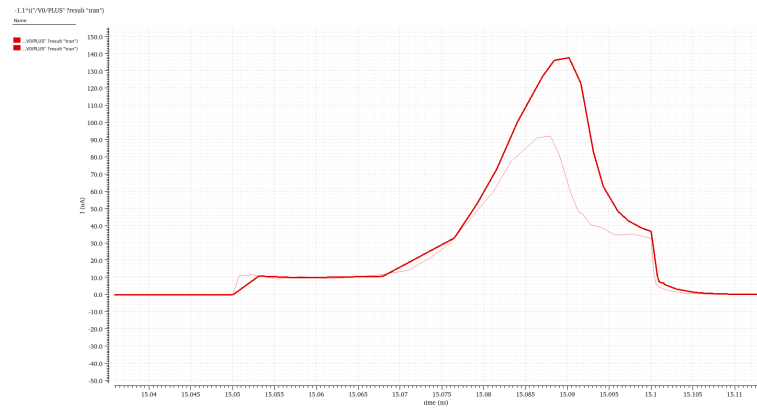


Figure 2.10: Power Consumption Plot of INV_X1

The above plot shows the instantaneous power consumption of the INV_X1 standard cell during switching operation. The total average power is calculated from the average current drawn from the VDD supply during transient simulation.

2.4.6 Static Power - Pre Layout

Input Condition	Static Leakage Power
A = 0	100.3 pW
A = 1	8.942 pW

Table 2.1: Static Leakage Power Measurement

2.4.7 Static Power - Post Layout

Input Condition	Static Leakage Power
A = 0	100.3pW
A = 1	8.942pW

Table 2.2: Static Leakage Power Measurement

2.4.8 Comparision Table

Parameter	Pre-layout	Post-layout Extracted	Delta	Observation
Rise time 10%-90%	9.051ps	12.21ps	3.159ps	Increased due to parasitics
Fall time 90%-10%	10.15ps	14.62ps	4.47ps	Slight increase after extraction
Rise delay t _{pLH}	9.841ps	13.38ps	3.539ps	Delay increased due to RC effects
Fall delay t _{pHL}	13.66ps	18.06ps	4.4ps	Output fall slowed after PEX
Average propagation delay	11.7505ps	15.72ps	3.9695ps	Overall delay increased
Total average power	110.5nW	237nW	126.5nW	Power increased after extraction
Static/leakage power (avg)	54.621pW	54.622pW	0.001pW	Minor leakage increase observed
Estimated dynamic power	110.445nW	236.945nW	126.5nW	Dynamic power increased
Cell area	2.088um ²	2.088um ²	0	No change in layout area

Chapter 3

OAI21 - X1

3.1 Cell Definition

Item	Specification
Cell name	OAI21_X1
Logic function	$Y = \overline{(A + B)C}$
Drive strength	X1
Technology	GPDK45nm
Transistor Sizing	PMOS : W = 1.35 μm , NMOS : W = 0.45 μm , Lmin = 45 nm
Template	9T standard cell, 1.8 μm height
Required pins	A, B, C, Y, VDD, VSS

3.2 Design View Created

3.2.1 Schematic and Symbol

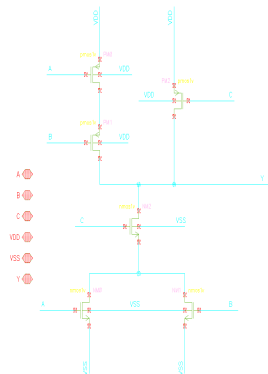


Figure 3.1: OAI21-X1-Schematic

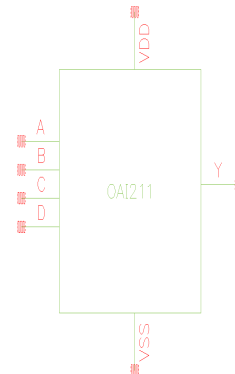


Figure 3.2: OAI21-X1-Symbol

3.2.2 Layout

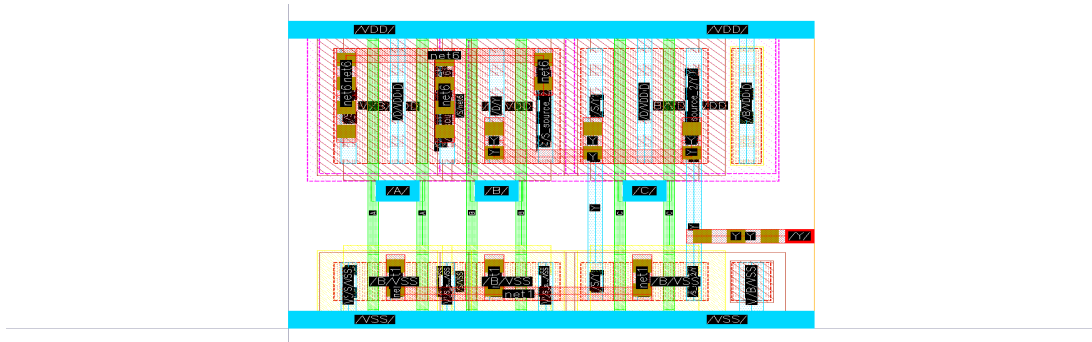


Figure 3.3: OAI21-X1-Layout

3.2.3 AV-Extracted layout

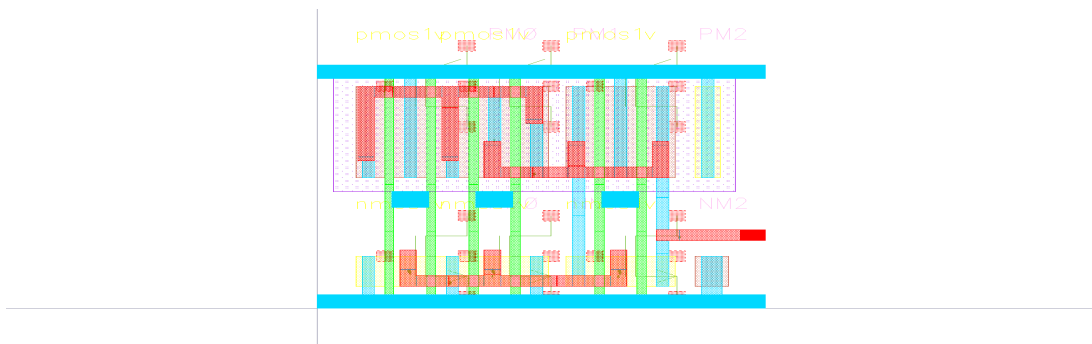


Figure 3.4: OAI21-X1-AV-Extracted

3.2.4 Testbench

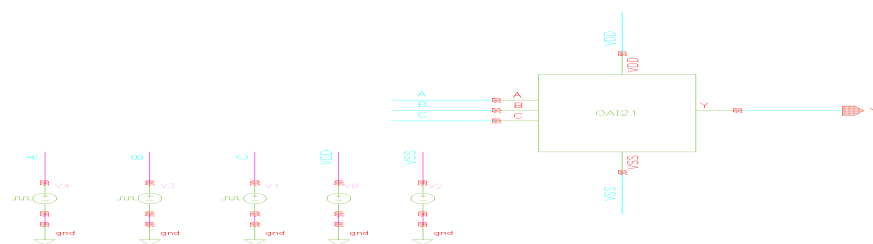


Figure 3.5: OAI21-X1-Testbench

3.3 Results

3.3.1 Static Power Comparison

Input Condition	Pre-Layout Static Leakage Power	Post-Layout Static Leakage Power
A, B, C = 0	255.8 pW	256.5 pW
A, B, C = 1	12.2 pW	12.13 pW

Table 3.1: Comparison of Pre-Layout and Post-Layout Static Leakage Power

3.3.2 Comparison Table

Parameter	Pre-layout	Post-layout Extracted	Delta	Observation
Rise time 10%-90%	14.14ps	22.68ps	8.54ps	Increased due to parasitics
Fall time 90%-10%	23.02ps	39.48ps	16.66ps	Slight increase after extraction
Rise delay t _{pLH}	21.36ps	28.18ps	6.82ps	Delay increased due to RC effects
Fall delay t _{pHL}	32.91ps	45.15ps	12.24ps	Output fall slowed after PEX
Average propagation delay	27.135ps	36.665ps	9.53ps	Overall delay increased
Total average power	262.5nW	490nW	227.5nW	Power increased after extraction
Static/leakage power (avg)	134pW	134.315pW	0.315pW	Minor leakage increase observed
Estimated dynamic power	262.36nW	489.866nW	227.5nW	Dynamic power increased
Cell area	3.93um ²	3.93um ²	0	No change in layout area

Chapter 4

OAI211 - X1

4.1 Cell Definition

Item	Specification
Cell name	OAI211_X1
Logic function	$Y = \overline{(A + B)CD}$
Drive strength	X1
Technology	GPDK45nm
Transistor Sizing	PMOS : W = 1.35 μm , NMOS : W = 0.45 μm , Lmin = 45 nm
Template	9T standard cell, 1.8 um height
Required pins	A, B, C, D, Y, VDD, VSS

4.2 Design View Created

4.2.1 Schematic and Symbol

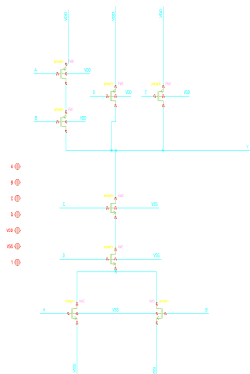


Figure 4.1: OAI211-X1-Schematic

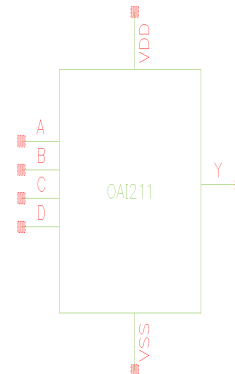


Figure 4.2: OAI211-X1-Symbol

4.2.2 Layout

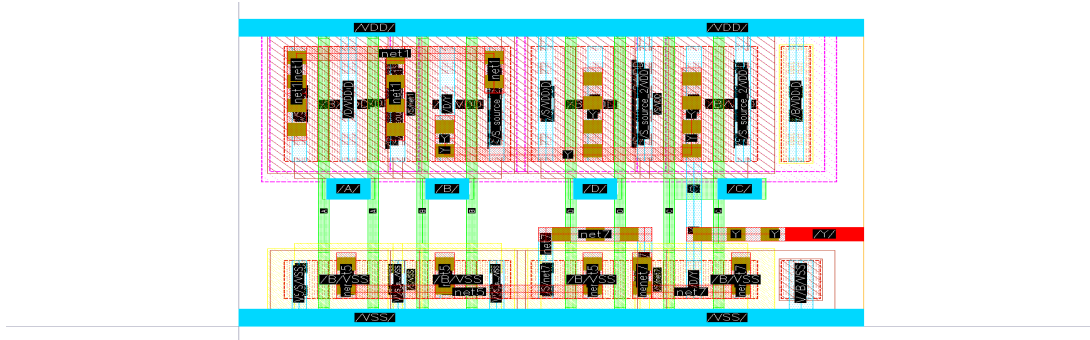


Figure 4.3: OAI211-X1-Layout

4.2.3 AV-Extracted layout

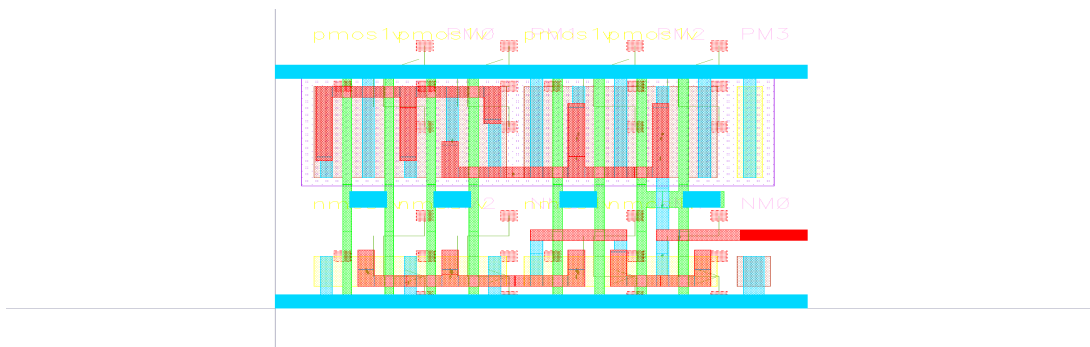


Figure 4.4: OAI211-X1-AV-Extracted

4.2.4 Testbench

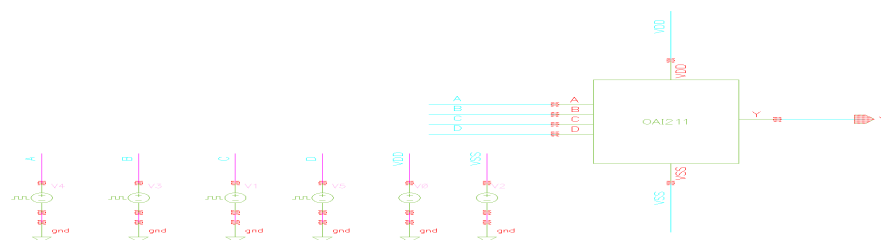


Figure 4.5: OAI211-X1-Testbench

4.3 Results

4.3.1 Static Power Comparison

Input Condition	Pre-Layout Static Leakage Power	Post-Layout Static Leakage Power
A,B,C,D = 0	335.7 pW	336pW
A,B,C,D = 1	21.14 pW	20.18 pW

4.3.2 Comparision Table

Parameter	Pre-layout	Post-layout Extracted	Delta	Observation
Rise time 10%-90%	18.47ps	30.28ps	11.81ps	Increased due to parasitics
Fall time 90%-10%	45.56ps	69.73ps	24.17ps	Slight increase after extraction
Rise delay t _{pLH}	23.9ps	31.9ps	8ps	Delay increased due to RC effects
Fall delay t _{pHL}	49.8ps	71.54ps	21.74ps	Output fall slowed after PEX
Average propagation delay	36.91ps	51.73ps	14.82ps	Overall delay increased
Total average power	386.6nW	702.1nW	315.5nW	Power increased after extraction
Static/leakage power (avg)	178.4pW	178.8pW	0.4pW	Minor leakage increase observed
Estimated dynamic power	385.422nW	701.9nW	316.478nW	Dynamic power increased
Cell area	4.671um ²	4.671um ²	0	No change in layout area

Chapter 5

OAI22 - X1

5.1 Cell Definition

Item	Specification
Cell name	OAI22_X1
Logic function	$Y = \overline{(A + B)(C + D)}$
Drive strength	X1
Technology	GPDK45nm
Transistor Sizing	PMOS : W = 1.35 μm , NMOS : W = 0.45 μm , Lmin = 45 nm
Template	9T standard cell, 1.8 μm height
Required pins	A, B, C, D, Y, VDD, VSS

5.2 Design View Created

5.2.1 Schematic and Symbol

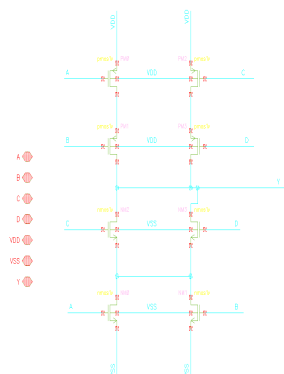


Figure 5.1: OAI22-X1-Schematic

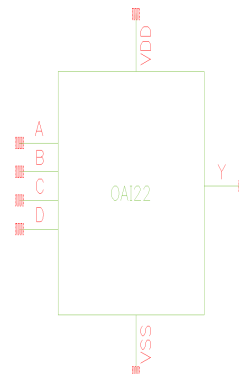


Figure 5.2: OAI22-X1-Symbol

5.2.2 Layout

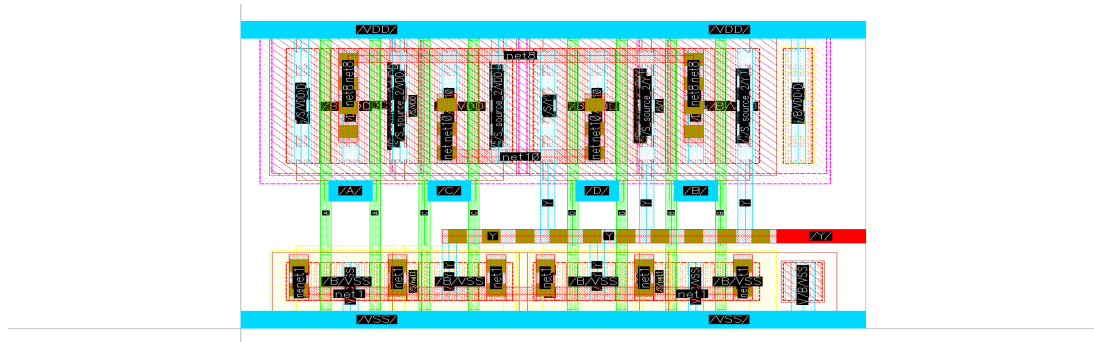


Figure 5.3: OAI22-X1-Layout

5.2.3 AV-Extracted layout

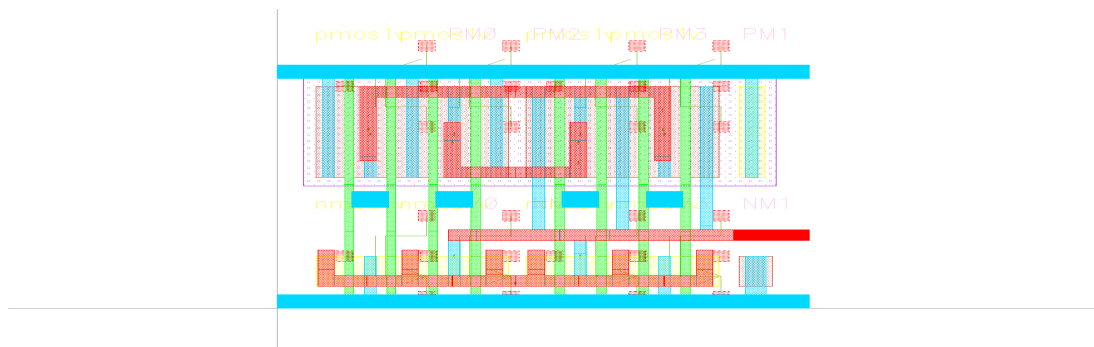


Figure 5.4: OAI22-X1-AV-Extracted

5.2.4 Testbench

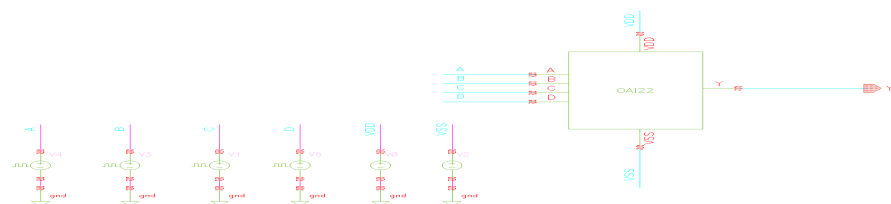


Figure 5.5: OAI22-X1-Testbench

5.3 Results

5.3.1 Static Power Comparison

Input Condition	Pre-Layout Static Leakage Power	Post-Layout Static Leakage Power
A,B,C,D = 0	343.3 pW	344.7 pW
A,B,C,D = 1	6.525 pW	6.342 pW

5.3.2 Comparison Table

Parameter	Pre-layout	Post-layout Extracted	Delta	Observation
Rise time 10%-90%	15.89ps	28.29ps	12.4ps	Increased due to parasitics
Fall time 90%-10%	20.9pps	35.64ps	14.74ps	Slight increase after extraction
Rise delay t _{pLH}	23.43ps	32.28ps	8.85ps	Delay increased due to RC effects
Fall delay t _{pHL}	31.48ps	42.72ps	11.24ps	Output fall slowed after PEX
Average propagation delay	27.45ps	37.5ps	10.05ps	Overall delay increased
Total average power	526.5nW	876.4nW	351.2nW	Power increased after extraction
Static/leakage power (avg)	174.9pW	175.5pW	0.6pW	Minor leakage increase observed
Estimated dynamic power	525.826nW	876.225nW	350.2nW	Dynamic power increased
Cell area	4.671um ²	4.671um ²	0	No change in layout area

Chapter 6

OAI221 - X1

6.1 Cell Definition

Item	Specification
Cell name	OAI221_X1
Logic function	$Y = \overline{(A + B)(C + D)E}$
Drive strength	X1
Technology	GPDK45nm
Transistor Sizing	PMOS : W = 1.35 μm , NMOS : W = 0.45 μm , Lmin = 45 nm
Template	9T standard cell, 1.8 um height
Required pins	A, B, C, D, E, Y, VDD, VSS

6.2 Design View Created

6.2.1 Schematic and Symbol

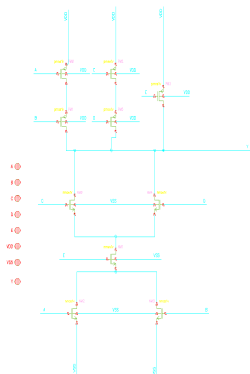


Figure 6.1: OAI221-X1-Schematic

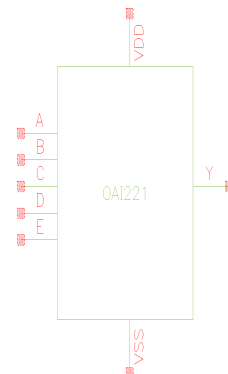


Figure 6.2: OAI221-X1-Symbol

6.2.2 Layout

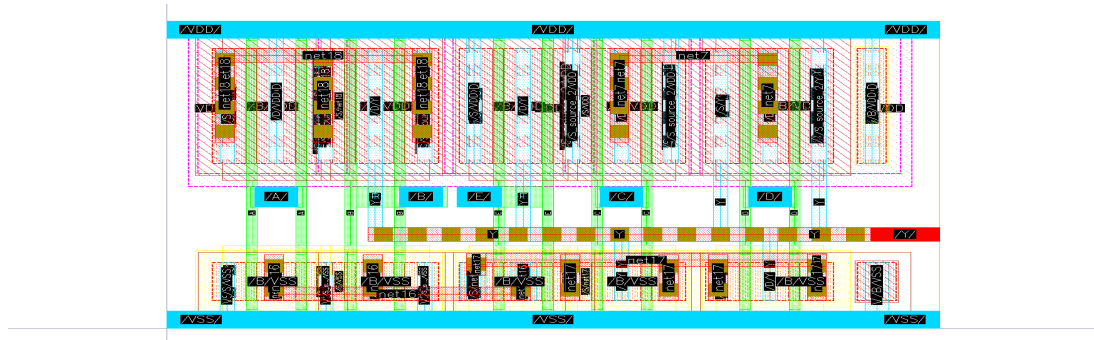


Figure 6.3: OAI221-X1-Layout

6.2.3 AV-Extracted layout

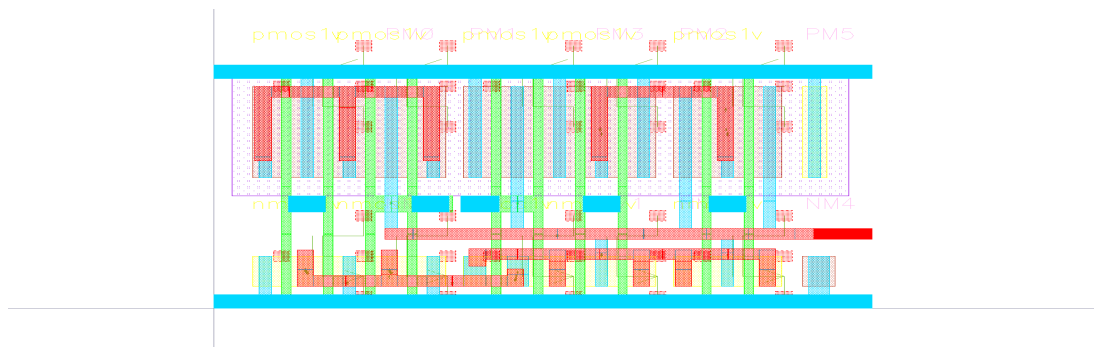


Figure 6.4: OAI221-X1-AV-Extracted

6.2.4 Testbench

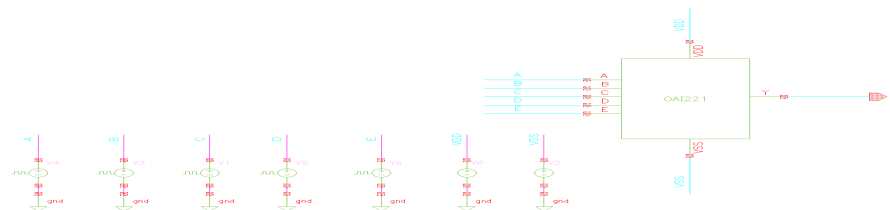


Figure 6.5: OAI221-X1-Testbench

6.3 Results

6.3.1 Static Power Comparison

Input Condition	Pre-Layout Static Leakage Power	Post-Layout Static Leakage Power
A,B,C,D,E = 0	422.6 pW	422.8 pW
A,B,C,D,E = 1	15.46 pW	15.07 pW

6.3.2 Comparison Table

Parameter	Pre-layout	Post-layout Extracted	Delta	Observation
Rise time 10%-90%	20.32ps	37.59ps	17.27ps	Increased due to parasitics
Fall time 90%-10%	57.89ps	102.4ps	44.51ps	Slight increase after extraction
Rise delay t _{pLH}	26.06ps	37.32ps	11.26ps	Delay increased due to RC effects
Fall delay t _{pHL}	66.2ps	99.25ps	33.05ps	Output fall slowed after PEX
Average propagation delay	46.13ps	68.285ps	22.155ps	Overall delay increased
Total average power	458.69nW	895.9nW	437.21nW	Power increased after extraction
Static/leakage power (avg)	219.03pW	218.935pW	-0.095pW	Minor leakage increase observed
Estimated dynamic power	458.171nW	895.582nW	437.411nW	Dynamic power increased
Cell area	5.778um ²	5.778um ²	0	No change in layout area

Chapter 7

Conclusions and future scope

7.1 Conclusion

This internship project successfully demonstrated the design, implementation, characterization, and physical verification of a standard cell library using GPD45nm technology in Cadence Virtuoso. The standard cells INV_X1, OAI21_X1, OAI211_X1, OAI22_X1, and OAI221_X1 were designed following a 9-track architecture and verified using Cadence Assura through DRC, LVS, and PEX flows. Pre-layout and post-layout simulations were performed to evaluate timing, power, and area characteristics, and the impact of layout parasitics on circuit performance was analyzed. All cells were found to be functionally correct, DRC-clean, and LVS-matched. This work provided valuable hands-on experience in custom VLSI design, physical verification, layout optimization, and standard cell characterization, while establishing a strong foundation for future ASIC and digital VLSI design applications.

7.2 Future Scope

The standard cell library developed in this work can be further extended by designing and characterizing more complex combinational and sequential cells such as multiplexers, encoders, decoders, latches, flip-flops, and arithmetic building blocks. Future work may also include the development of cells with multiple drive strengths (X2, X4, and X8) to support different timing and load requirements in digital ASIC designs. Further characterization can be performed across different process, voltage, and temperature (PVT) corners to evaluate cell robustness and reliability under varying operating conditions. The generated standard cells can also be characterized for timing and power using industry-standard characterization tools to generate Liberty (.lib) files, LEF files, and other views required for complete digital design flows. In addition, layout optimization techniques can be explored to reduce area, delay, and power consumption while improving routing accessibility and manufacturability. The methodology developed in this project can serve as a foundation for creating a complete standard cell library suitable for advanced VLSI and ASIC design applications.

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